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ORIGINAL PAPER

The diagnostic validity of central symptoms for major depressive disorder in adults with attention-deficit/hyperactivity disorder: a network analysis

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Abstract

Background An efficient and accurate diagnosis of MDD in adult ADHD is highly significant. Central symptoms identified through network analysis are considered to play a key role in diagnosis, potentially contributing to the achievement of this goal.

Methods ADHD and MDD symptoms were assessed using the ADHD Rating Scale and Zung Self-Rating Depression Scale (SDS) in a clinical cohort of 418 adults with ADHD, followed by network estimations. Symptoms with high expected influence (EI) in the network were identified as central symptoms and tested for diagnostic validity via ROC curve analysis. Imaging analyses were further adopted to verify the effectiveness of the identified central symptoms from the perspective of construct validity.

Results A five-symptom model was identified, including: *"Depressed affect," "Personal devaluation," "Emptiness," "Agitation,"* and *"Interest loss."* The model exhibited strong diagnostic validity for MDD in adult ADHD, showing no significant difference compared to using all 20 SDS items. Moreover, significant functional alterations were detected in the left anterior cingulate cortex between groups classified by this model.

Conclusion A five-symptom model comprising *"Depressed affect," "Personal devaluation," "Emptiness," "Agitation,"* and *"Interest loss"* showed strong diagnostic validity for MDD in adult ADHD, and proven effective in exploring its underlying neural mechanisms. These symptoms should be considered a priority in diagnosing MDD in adults with ADHD.

Keywords Attention-deficit/hyperactivity disorder · Major depressive disorder · Network analysis · Diagnostic validity

Introduction

Attention-deficit/hyperactivity disorder (ADHD) is characterized by age-inappropriate levels of inattention, hyperactivity, and impulsivity, with a global prevalence of 2.5% in adults [1]. Major depressive disorder (MDD) is one of its most common comorbidities, with adults with ADHD being three times more likely to experience MDD compared to the general population [2]. MDD in the ADHD population is often characterized by earlier onset, greater chronicity, and higher rates of self-harm and suicide [3]. Moreover, comorbid MDD is associated with more severe functional impairments and worse treatment outcomes in adults with ADHD [4, 5]. Therefore, an early and accurate MDD diagnosis in ADHD is crucial.

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However, screening for MDD in ADHD can be challenging, partly due to the limitations of current diagnostic tools, which rely on standardized items with equal weighting [6]. These tools assume that all patients with MDD experience depressive symptoms of equal significance, disregarding the possibility that certain symptoms may be more salient for some individuals. This approach fails to account for the heterogeneity of MDD and does not consider the additional impact of comorbid conditions like ADHD [7].

The network approach offers a promising solution to this issue. It conceptualizes disorders as systems of interconnected symptoms, with the relative importance of each symptom quantified through centrality indices [8, 9]. Symptoms with high centrality indices are considered to have a strong influence on other symptoms or to contribute significantly to comorbid conditions [10]. As such, these symptoms should be prioritized in diagnosis [6].

Over the past decade, network analysis has been widely applied to MDD, yielding many clinically significant findings [11]. However, studies specifically focusing on MDD in adults with ADHD remain scarce, despite the ongoing challenges in diagnosis mentioned above. In this study, we aim to identify which depressive symptoms are most central to the diagnosis of MDD in adults with ADHD, while also exploring the ability of these symptoms to reveal underlying neural mechanisms, in order to further validate their diagnostic validity.

Materials and methods

Participants

A total of 418 participants were recruited from Peking University Sixth Hospital. Inclusion criteria were: (1) a confirmed diagnosis of adult ADHD diagnosis in accordance with the fourth edition of the American Psychiatric Association' s Diagnostic and Statistical Manual of Mental Disorders (DSM-IV; American Psychiatric Association [APA], 1994) [12], (2) age 18–45, (3) drug-naïve or off medication for half a year, (4) full-scale IQ >80. Exclusion criteria included: history of schizophrenia, pervasive developmental disorders, significant brain trauma, or suicidal ideation/planning/attempt. All participants provided written informed consents. The study was approved by the Ethics Committee of Peking University Institute of Mental Health.

Assessments

The diagnosis of adult ADHD was made by trained psychiatrists using the Conner's Adult ADHD Diagnostic Interview (CAADID) [13]. Comorbidities were assessed via the Structural Clinical Interview for DSM-IV Axis I Disorders (SCID-I) [14]. Participants also completed the following assessments: (1) the Wechsler Adult Intelligence Scale-Revised in China, Third Edition (WAIS-RC), to obtain a full-scale IQ score; (2) the ADHD Rating Scale (ADHD-RS) is a 4-point Likert-type scale (1 = not at all, 4 = always) comprising 9 items assessing Inattention and 9 items assessing Hyperactivity/Impulsivity, used to evaluate the severity of core ADHD symptoms [15]; and (3) the Self-Rating Depression Scale (SDS) is a 20-item self-report scale (1 = not at all, 4 = most of the time) used to assess symptoms of MDD, with higher scores indicating more severe depressive symptoms.

MRI data acquisition and preprocessing

A subgroup of 230 participants underwent magnetic resonance imaging (MRI) scanning, with exclusion criteria included mental implants, claustrophobia, and other conditions unsuitable for scanning. Resting-state functional MRI (rs-fMRI) and high-resolution T1-weighted anatomical imaging were acquired using a 3T MR scanner (General Electric; Discovery MR750). During the scan, participants were instructed to remain motionless, eyes closed, and awake. After excluding participants with head movement exceeding 3.0 mm in translation or 3 degrees in rotation, 223 participants were included in further analysis.

MRI data were processed using the RESTplus toolbox [16] and SPM12 software (https://www.fil.ion.ucl.ac.uk/spm/) on MATLAB R2013b. The amplitude of low-frequency fluctuations (ALFF) and regional homogeneity (ReHo) were extracted, followed by Fisher's z transformation and smoothing. Detailed parameters and processing steps are provided in the supplementary materials.

Network analysis

The Gaussian Graphical Model (GGM) was used to estimate an undirected network, where nodes represent symptoms and edges depict their conditional associations. The edge thickness indicates the magnitude of these associations, with green and red representing positive and negative associations, respectively. Regularization was applied to minimize the risk of false edges. The Graphical Least Absolute Shrinkage and Selection Operator (gLASSO) and the Extended Bayesian Information Criteria (EBIC) were used for model selection to ensure a relatively stable network. Network visualization was performed using the *"qgraph"* package [17].

Referred to previous studies, Expected Influence (EI) was used to identify central symptoms, as it provides greater

stability and accounts for negative edges in the network [18, 19].

Network accuracy and stability were assessed using the *"bootnet"* package with 5,000 bootstrap samples [20]. During this process, non-parametric Confidence Intervals (CIs) and the correlation stability (CS) coefficients were calculated to evaluate edge accuracy and centrality stability, respectively.

Sensitivity analysis

To eliminate the potential influence of other comorbidities, we reconstructed the symptom network and explored the central symptoms after excluding participants with comorbid bipolar disorder (BD) or anxiety disorders.

Comparison of diagnostic validity

The twenty depressive symptoms in the SDS were sequenced based on descending and ascending EI values, with each item added sequentially to form cumulative sums for both orders. These cumulative sums for both the descending and ascending EI groups were then used to predict clinically diagnosed MDD in adults with ADHD. Receiver operating characteristic (ROC) curves were plotted, and the area under the curve (AUC) along with the Youden index were calculated to evaluate diagnostic validity. This comprehensive analysis allowed us to observe how diagnostic performance changed as symptoms were added, thereby directly illustrating the clinical significance of central symptoms. Additionally, we compared the diagnostic validity of a subset of central symptoms against the full set of all twenty symptoms. This comparison across different cutoff points was conducted to further demonstrate that a small number of central symptoms could achieve diagnostic performance on par with the full symptom list. In this step, the bootstrap function in the R package "cutpointr" [21] was employed to ensure the robustness and stability of the results.

Table 1 Description of demographic and clinical characteristics (N=418)

Variables	Mean(SD)/counts(%)
Sex	
Male	260(62.2)
Female	158(37.8)
Age	26.54(5.36)
FSIQ	120.42(9.65)
Comorbidity	
MDD	52(12.4)
BD	16(3.8)
Anxiety disorder	59(14.1)
ADHD total score	28.42±8.17
Inattention	17.95±4.26
Hyperactivity/impulsivity	10.47±5.66
SDS total score	40.63±9.69

FSIQ full-scaled intelligence quotient, *ADHD* attention-deficit/hyperactivity disorder, *MDD* major depressive disorder, *BD* bipolar disorder, *SDS* self-rating depression scale, *SD* standard deviation

Neuroimaging analyses

Based on the optimal cutoff point for central symptom subset, participants were divided into two groups as ADHD + $_{central}$MDD (above the cutoff point) and ADHD-$_{central}$MDD (below the cutoff point). Controlling for age, sex, and head motion, ALFF and ReHo values were compared between the two groups. Multiple comparisons were corrected using Gaussian Random Field (GRF) theory at a voxel-level threshold of $p < 0.001$ and a cluster-level threshold of $p < 0.05$ [22].Subsequently, signals were extracted from the significant clusters, and their correlations with SDS T-scores and central symptom scores were analyzed to confirm the associations.

Results

Demographic and clinical characteristics

Demographic and clinical characteristics of the study population are presented in Table 1. A total of 418 adult participants with ADHD were recruited, of whom 62.2% were male, with a mean age of 26.54±5.36 years. There were 52 (12.4%) participants also diagnosed with MDD.

Symptom network and centrality plot

Figure 1A shows the network structure of ADHD and depressive symptoms, with the correlation matrix shown in Figure S1. For the whole network, the strongest associations were between HI2-HI3 (*"Seat"* - *"Restless"*, $r = 0.34$) and IA4-IA5 (*"Follow"* - *"Organize"*, $r = 0.34$). For depressive symptoms, the strongest associations were between D14-D17 (*"Hopelessness"- "Personal devaluation"*, $r = 0.27$) and D1-D18 (*"Depressed affect"- "Emptiness"*, $r = 0.24$). The centrality plot (Fig. 1B) displayed the EI values of these symptoms. D1 had the highest EI value, suggesting it is the most central symptom in the whole network. Other depressive symptoms with high EI values were as follows: D1, D17, D18, D13 (*"Agitation"*), and D20 (*"Interest loss"*). These symptoms maintained the same ranking in the sensitivity analysis. The network demonstrated good stability and accuracy (presented in Figure S2 and 3).Codes for symptoms were shown in Table S1.

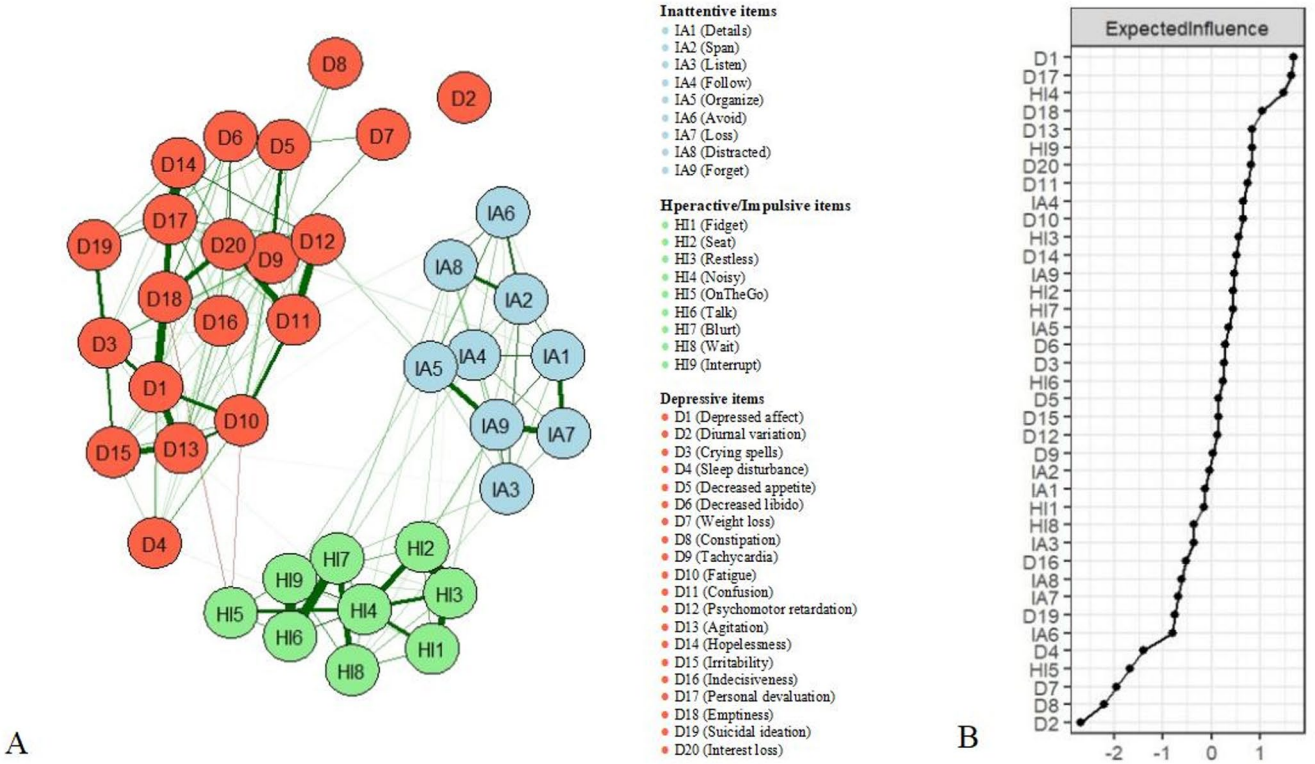


Fig. 1 Network of ADHD and depression symptoms in adults with ADHD. **A** Network nodes and edges plot. **B** ExpectedInfluence plot (z-score). Notes: red nodes indicate depression symptoms; blue nodes indicate inattentive symptoms; green nodes indicate inattentive symptoms. IA1-IA9 represented 9 inattentive symptoms in ADHD; HI1-H9 represented 9 hyperactive/impulsive symptoms in ADHD; D1-D20 represented 20 depressive symptoms from SDS

Comparison of diagnostic validity

Forty ROC curves were plotted for two groups based on the diagnostic models. The AUC and Youden indices, shown in Fig. 2, revealed that central symptoms had significantly higher diagnostic validity than marginal symptoms. When only one symptom was included in the model, the AUC for central symptoms was 0.812, whereas for marginal symptoms it was 0.481 (z = 6.02, $P < 0.001$). This performance gap gradually narrowed as more symptoms were added. The difference in AUC between central and marginal symptoms became statistically non-significant only after 13 symptoms were included (z = 1.55, $P = 0.121$). The Youden indices exhibited a similar trend. For central symptoms, the first four yielded diagnostic validity comparable to all 20 SDS items (z = 1.61, $P = 0.107$). Notably, the model containing top 5 central symptoms yielded the highest diagnostic validity among small symptom sets, forming the first peak in the ROC trajectory (as observed in Fig. 2). This finding suggested a model with top 5 central symptoms might offer an optimal balance between brevity and predictive validity. Mapping of these symptoms to the DSM-5 diagnostic criteria was shown in Table 2.

As shown in Fig. 3, for the five central symptoms, a cut-off of 14 yielded the highest AUC of 0.856, with a Youden index of 0.671, sensitivity of 0.846, and specificity of 0.825. For all twenty SDS items, the optimal cut-off was 56, yielding an AUC of 0.863, sensitivity of 0.846, and specificity of 0.746. There was no significant difference in diagnostic validity between using the central symptoms or all twenty symptoms at their respective optimal cut-offs (z = -0.45, $P = 0.650$), even with a higher specificity for central symptoms (0.825 vs. 0.746). The stability of these findings was confirmed by consistent cutoff points and AUCs from bootstrap sampling (Figure S4).

Association with neural mechanism

Under the optimal cutoff point, the 5-central-symptom model classified participants into two groups: the "ADHD + $_{central}$MDD" group ($n = 108$, nMRI = 49) and the "ADHD– $_{central}$MDD" group ($n = 310$, nMRI = 174). Between-group comparisons revealed significantly higher ALFF and ReHo values in the left ACC (Fig. 4A and B). After controlling for age, sex and head motions, both ALFF and ReHo in the left ACC showed significant positive correlations with the the total score of these 5 central symptoms

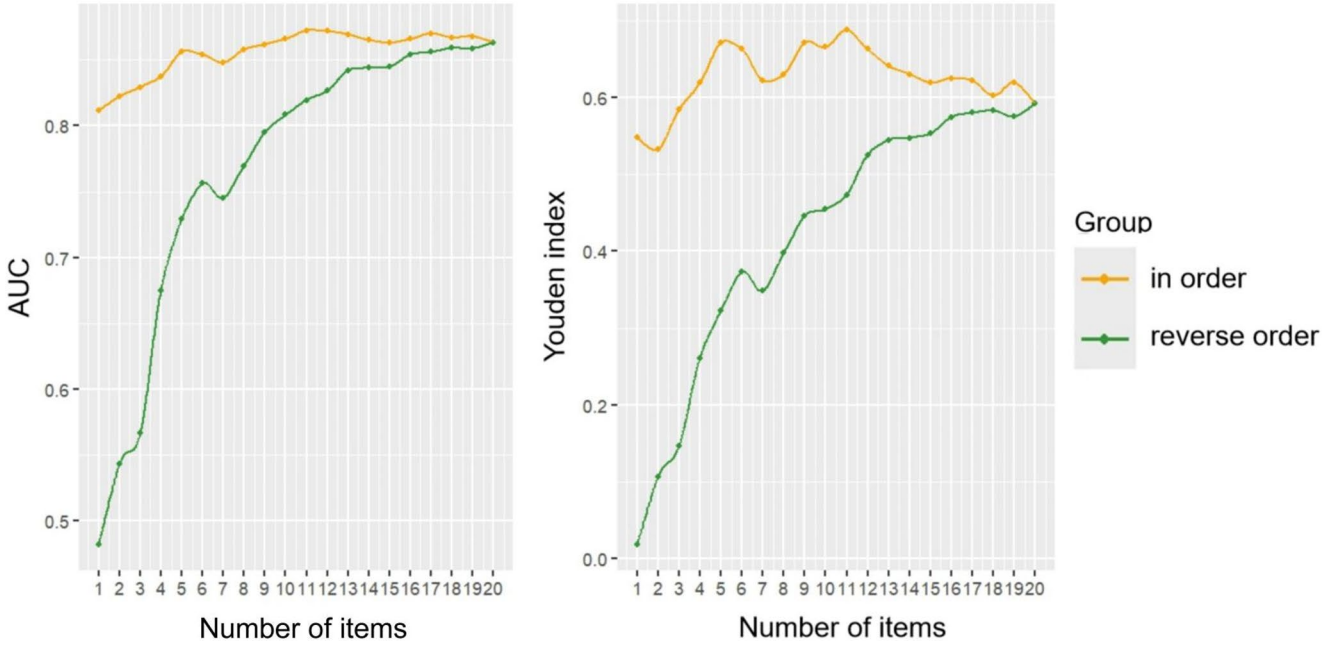


Fig. 2 Changes of auc and Youden index of diagnostic ROC with SDS items included in the model increasing one by one. The orange group (in order) bring SDS items into the model following the descending order of EI values. The green group (reverse order) following the ascending order

Table 2 Mapping of central symptoms to DSM-5 criteria and their centrality ranking in other contexts

Central symptoms	Description in the DSM	adultADHD	MDD [29]	BD [29]	Community samples [28]
Depressed affect	Feeling down, depressed, or hopeless.	1	1	1	1
Personal devaluation	Feeling bad about yourself, or that you are a failure or have let yourself or your family down.	2	4	5	7
Emptiness	–	3	5	2	–
Agitation	Being so fidgety or restless that you have been moving around a lot more than usual.	4	9	8	13
Interest loss	Little interest or pleasure in doing things.	5	10	14	3

(ALFF: $r=0.17$, $p=0.009$; ReHo: $r=0.20$, $p=0.003$; Fig. 4C). Additionally, when participants were categorized using the optimal cutoff point for all 20 depressive symptoms (T-score ≥ 56), significantly increased ALFF and ReHo values were observed in the left ACC in the higher-score group (ALFF: $p=0.004$; ReHo: $p=0.008$), and these values were also positively correlated with the total score of all 20 depressive symptoms (ALFF: $r=0.13$, $p=0.048$; ReHo: $r=0.19$, $p=0.003$; Fig. 4C).

Discussion

This study investigated the potential of central depressive symptoms for diagnosing MDD in adults with ADHD. A set of five central depressive symptoms were identified, including: *"Depressed affect," "Personal devaluation," "Emptiness," "Agitation," and "Interest loss."* These symptoms demonstrated better diagnostic validity than marginal symptoms and as effective as all SDS items, with even higher specificity. Significant functional alterations in the left anterior cingulate were observed between the two groups defined by central symptoms. These findings suggest that these central symptoms could serve as potential indicators for diagnosing MDD in adults with ADHD.

This study constructed a symptom network of 38 nodes, comprising 18 ADHD symptoms and 20 depressive symptoms. Depressive symptoms had generally higher EI values than ADHD symptoms, indicating their stronger influence

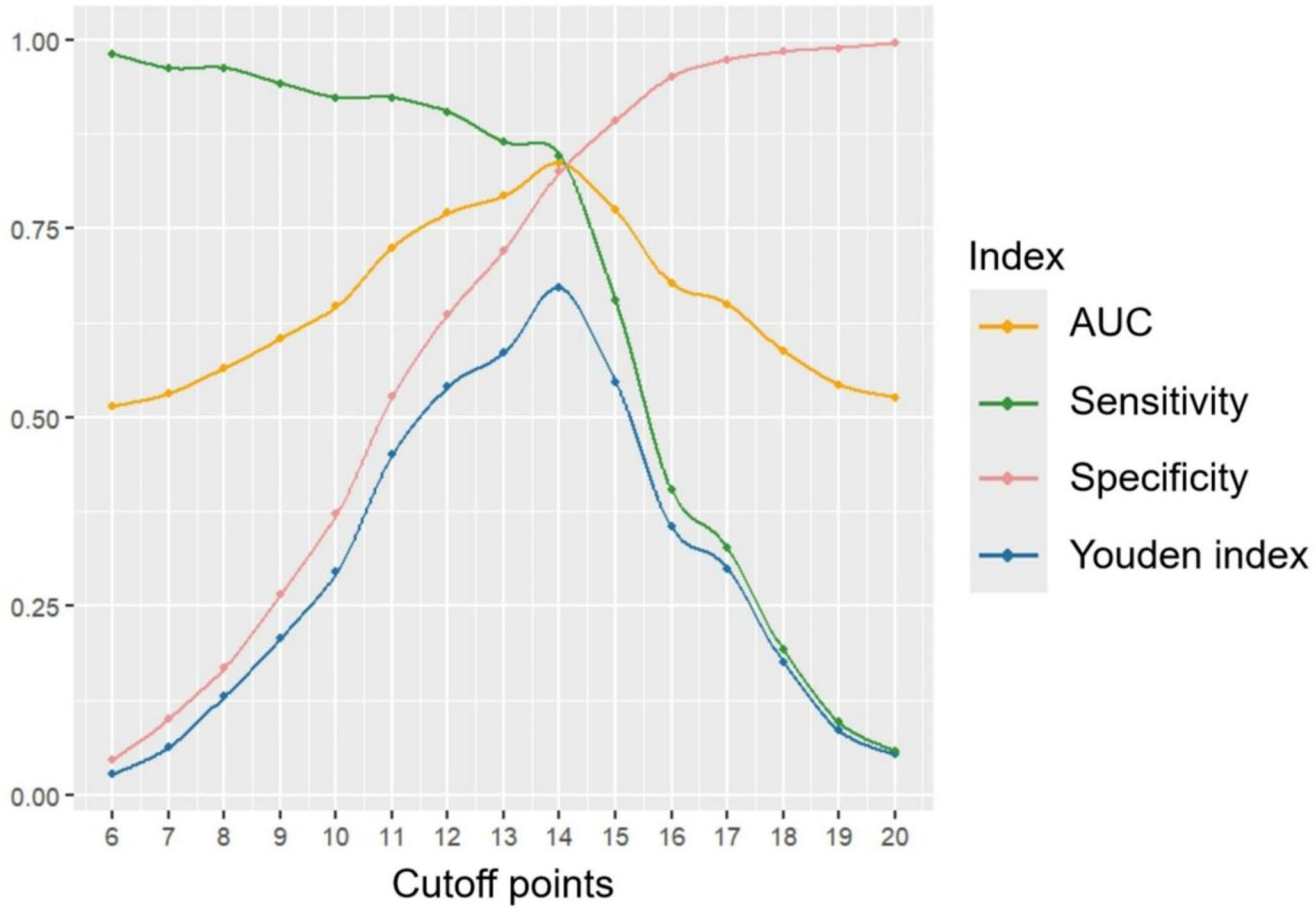


Fig. 3 The diagnostic ROC indices under different cutoff of central symptoms. Orange indicates AUC; green indicates sensitivity; pink indicates specificity; blue indicated Youden index. The values of auc and Youden index are largest when cutoff is 14

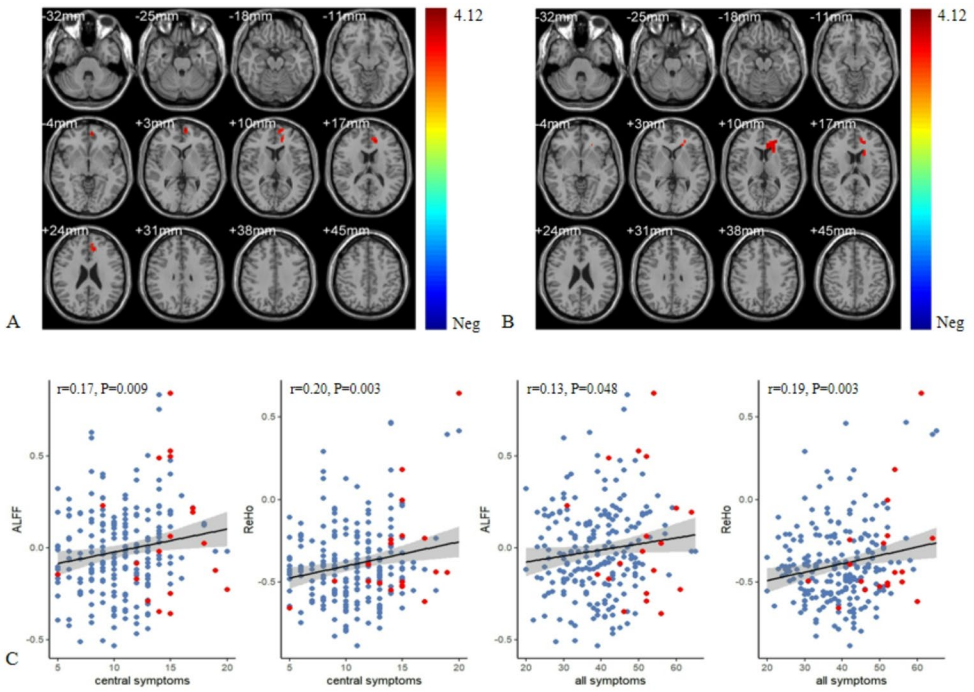


Fig. 4 Brain regions showing significantly altered ALFF and ReHo between ADHD + $_{central}$MDD and ADHD-$_{central}$MDD, as well as their association with clinical symptoms. **A** ALFF **B** ReHo **C** the correlation diagram. (GRF correction, voxel-level $p < 0.005$, cluster-level $p < 0.05$). The color bar in **A** and **B** indicated the t value. Warm colors indicated increased indicator values, while cold colors indicated decreased indicator values. The red nodes in (C) meant participants in ADHD + $_{central}$MDD group, while blue meant in ADHD-$_{central}$MDD group. *ALFF* amplitude of low-frequency fluctuation, *ReHo* regional homogeneity

on other symptoms within the network [10]. This meant that depressive symptoms, rather than ADHD symptoms, might play a more key role when ADHD and MDD co-occur. A possible explanation is that worsened emotional issues following the comorbidity of MDD may significantly affect attention, memory, and decision-making, thereby leading to greater functional impairments [23, 24].

The five symptoms are also commonly observed in other mood disorders in adult ADHD, such as BD and anxiety disorders [25, 26]. Considering the potential effect of the comorbid status of these disorders, we have excluded those with comorbidities and repeated the analyses. The results were retained, supporting the core role of these five symptoms for the identification of MDD in adults with ADHD. It should be emphasized that the values of these identified central symptoms do not lie in capturing manifestations unique to MDD, but rather in identifying a subset most strongly associated with and offering the greatest diagnostic contribution to this disorder [10, 27]. As Ebrahimi et al. suggested, compared to individual symptoms, symptom co-occurrence patterns may be more indicative for capturing the distinctive features of certain psychiatric disorder [11]. Of course, it is highly worthwhile to recruit subjects with different diagnosis (MDD, BD or anxiety disorders) to further elucidate the transdiagnostic nature of specific symptom or the diagnostically specific pattern of co-occurred symptoms, which will promote our understanding of the psychopathology of different mental disorders.

Among these central symptoms, *"Depressed affect"* exhibited the highest EI value. In community populations [28] and populations diagnosed with other mental disorders [29], although significant differences were observed in centrality rankings, *"Depressed affect"* was consistently the most central symptom. It is widely recognized as a hallmark clinical feature of MDD and serve as a core diagnostic criterion in the DSM [30]. This finding aligns with previous network studies, which have consistently identified *"Depressed affect," "depressive mood,"* or *"sadness"* as central symptoms across diverse populations, comorbidities and cultural contexts [31–33].

The strongest association with *"Depressed affect"* was observed for *"Emptiness." "Emptiness."* from the SDS, along with another central symptom *"Loss of interest,"* could reflect anhedonia [34], which is a core component of the DSM diagnostic criteria for MDD, as well as a necessary feature for its clinical diagnosis. Anhedonia plays an important role in the psychopathology of MDD [35, 36], serving as a indicator of disease severity and progression [37]. In addition, anhedonia is one of the most frequently overlapping symptoms between ADHD and MDD [38, 39], and has been identified in previous network analysis as a transdiagnostic trait linking the two disorders [40]. Interestingly, in the current study, *"Emptiness"* was negatively correlated with a representative ADHD symptom *"OnTheGo"* (i.e., act as driven by a motor). The presence of pronounced anhedonia can markedly reduce energy levels, even in the ADHD population. It is supported by previous research that adults with ADHD exhibit sustained energy and engagement primarily when participating in activities they find interesting or rewarding [41]. Conversely, this insight also implies that facilitating the identification of personally motivating activities may serve as an effective strategy for alleviating depressive symptoms in this population [42].

"Personal devaluation" was a central symptom, second only to *"Depressed affect."* Feelings of worthlessness are strongly associated with suicidal thoughts and attempts [43, 44]. In adults with ADHD, impaired executive functions make them more vulnerable to setbacks in school, work, and relationships [45]. When combined with depressive symptoms, these setbacks can amplify feelings of uselessness, which in turn exacerbate ADHD symptoms, making it more difficult to concentrate or complete tasks [46]. Our network analysis revealed direct connections between *"Personal devaluation"* and inattentive symptoms *"Detail"* and *"Organize"* in ADHD, which reflected difficulties in learning and work, thereby providing empirical support for this pathway. This growing sense of failure may contribute to the elevated suicide risk among individuals with ADHD [47]. As such, addressing *"Personal devaluation"* could be a key intervention target, particularly when ADHD co-occurs with MDD. Cognitive behavioral therapy (CBT) can help patients rebuild negative self-cognitions, break the cycle of failure, and improve both ADHD and depressive symptoms [48].

Another intriguing finding is the symptom of *"Agitation (i.e. feeling restless),"* which is commonly observed in MDD but less emphasized compared to other symptoms like sadness or anhedonia [6, 29]. In contrast, among individuals with ADHD, particularly adults, an analogous item *"feeling restless,"* represents a significant manifestation [49] and occupies a prominent position within the ADHD symptom network [50]. Consistent with this, Lee et al. identified *feeling restless* as the most central ADHD symptom in patients with mood disorders, while not in mood disorders alone [29, 51]. Collectively, these findings suggest that "*Agitation"* may be a distinctive symptom that warrants greater attention when MDD and ADHD co-occur, underscoring its significance for understanding the comorbidity and informing potential interventions.

As anticipated, above central symptoms demonstrated significantly higher diagnostic validity for MDD in adult ADHD compared to marginal symptoms. Moreover, a brief model consisting of the first five central depressive symptoms performed similarly to the full model, which included

all twenty SDS items, for diagnosing MDD in ADHD, even exhibiting greater specificity. This finding supports network theories that suggest central symptoms may be more clinically relevant and could serve as key targets for both diagnosis and intervention [10, 27]. The results highlight the potential benefits of using a streamlined set of central symptoms for clinical screening of MDD in ADHD, and potentially other psychiatric conditions. Central symptoms should be given greater weight in MDD evaluations in ADHD populations, as this could enhance diagnostic accuracy. Longitudinal studies could further validate the predictive significance of these central symptoms.

Previous studies have highlighted the importance of the ACC in MDD-related neural networks [52], with its dysfunction being a predictor of responses to antidepressant treatments and psychotherapy [53, 54]. Additionally, the ACC has also been implicated in ADHD [55]. Its central role in the reward system may explain why both ADHD and MDD are associated with damage to this brain region, suggesting the ACC could be a key neural mechanism underlying this comorbidity [56]. The observed differences in this well-validated biological marker further support the value of central symptoms in discriminating MDD in adults with ADHD. This symptom-level approach may also, to some extent, reflect its underlying neural mechanisms. The results support the feasibility of improving diagnosis and intervention through a focus on central symptoms.

Our study has several limitations. First, we were only able to construct an undirected network model base on the cross-sectional data. Future longitudinal studies are needed to explore the dynamic changes and potential causal relationships between ADHD and depressive symptoms, which would allow for a more clinically meaningful interpretation of central symptoms. Second, the symptom inclusion for the network was constrained by the limitations of the assessment tools. Although both the ADHD-RS and SDS are among the most widely used assessment tools in their respective domains, they do not capture certain relevant symptoms. For instance, Dong et al. proposed that emotional dysregulation might be a prominent feature distinguishing adults with comorbid ADHD and MDD from those with ADHD alone, yet neither of the scales used in this study includes relevant items [7]. Future studies should incorporate additional instruments to more comprehensively depict the symptom landscape of the disorders.

Conclusion

Our study identified central symptoms via network approach and innovatively demonstrated their diagnostic validity from both clinical and neurobiological perspectives. A model incorporating the top five central symptoms—"*Depressed affect*", *"Personal devaluation"*, *"Emptiness"*, *"Agitation"* and *"Interest loss"*—showed diagnostic validity comparable to that of the full symptom set. These findings suggest that particular attention should be paid to these central symptoms in optimizing the diagnostic process for MDD in adults with ADHD.

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Author contribution LL and QQ conceptualized, designed, and supervised the study. ZZ, XY, FS, MZ, MD, and HL contributed to experimental programming and data collection. LL, MP, YW and QQ contributed to the data interpretation. GY analyzed the data.YG and MP drafted and revised the manuscript. LL and QQ reviewed the manuscript and provided constructive suggestions. All authors have read and approved the final version of the manuscript for submission.

Data availability The data analysed in the current study are not publicly available, but are available from the corresponding author on reasonable request.

Declarations

Conflict of interest No potential conflict of interest was reported by the author(s).

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